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Multivariate Analysis for Multiple Network Data via Semi-Symmetric Tensor PCA

Michael Weylandt^a  and George Michailidis^b

^aZicklin School of Business, Baruch College, CUNY, New York, NY; ^bDepartment of Statistics & Data Science, University of California Los Angeles, Los Angeles, CA

ABSTRACT

Network data are commonly collected in a variety of applications, representing either directly measured or statistically inferred connections between subjects or features of interest. In an increasing number of domains, these networks are collected over time, such as repeated interactions between users of a social media platform, or across multiple subjects, such as in multi-subject neuroimaging studies. When analyzing multiple large networks, dimensionality reduction techniques are often used to embed networks in a more tractable low-dimensional space. To this end, we develop a framework for principal components analysis (PCA) on collections of networks via a specialized tensor decomposition, termed Semi-Symmetric Tensor PCA or SST-PCA, and analyze it theoretically. Notably, we show that SST-PCA achieves the same accuracy as classical matrix PCA, with error proportional to the square root of the number of vertices and not the number of edges as might be expected. Our framework inherits many of the strengths of classical PCA and is suitable for a wide range of unsupervised learning tasks, including identifying principal networks, isolating changepoints and outliers, and for characterizing the “variability network” of the most varying edges. Finally, we demonstrate the effectiveness of SST-PCA in simulation and on an example from empirical social studies. Supplementary materials for this article are available online, including a standardized description of the materials available for reproducing the work.

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CP factorization; Network analysis; Principal components analysis; Regularized PCA; Semi-symmetric tensor; Tensor decomposition

1. Introduction

Principal Components Analysis (PCA) is a fundamental tool for the analysis of multivariate data, enabling a wide range of dimension reduction, pattern recognition, and visualization strategies. Originally proposed for dimension reduction of numerical data in Euclidean space, PCA has been generalized to much more general contexts, including data taking values in general inner product spaces (Eaton 2007), (possibly infinite-dimensional) function spaces (Silverman 1996), discrete and compositional spaces (Michailidis and de Leeuw 1998; Liu, Dobriban, and Singer 2018), and shape spaces (Jung, Dryden, and Marron 2012). In this work, we analyze PCA-type decompositions for *network data*: that is, given a collection of *undirected* networks on a common set of nodes, we seek to identify common patterns in the form of one or more “principal networks” that capture most of the variation in the data. The main technical tool is a *tensor decomposition* on so-called *semi-symmetric tensors*, for which we provide an efficient computational algorithm and rigorous statistical guarantees, in the form of a general finite sample consistency result. As shown in the sequel, this approach achieves the same convergence rates as classical (Euclidean) PCA up to a logarithmic factor, with error proportional to the *square root* of the number of vertices rather than the number of edges, providing significant improvements when analyzing networks of even moderate size.

Network-valued data are of increasing importance in analyzing modern structured datasets. In many situations,

multiple networks are observed in conjunction with the same phenomenon, typically when independent replicates are observed over time (“network series”) or when different network representations of the same underlying dynamics can be observed (Kivlä et al. 2014, “multilayer networks”). Notable examples of this type of data can be found in neuroscience (Zhang et al. 2019), group social dynamics (Eagle, Pentland, and Lazer 2009), international development (Hafner-Burton, Kahler, and Montgomery 2009), and transportation studies (Cardillo et al. 2013). Because these undirected (weighted) networks are observed on a common known set of nodes, they can each be represented as symmetric matrices that can be stacked into a third-order *semi-symmetric* tensor, that is, a tensor each slice of which is a symmetric matrix. We study the decomposition depicted in Figure 1: that is, decomposing a semi-symmetric tensor into a “principal network” term and a loading vector. Details of this decomposition follow in Section 2.

The focus of the work is on scenarios wherein networks are directly observed and the structure of the networks is of primary interest. We distinguish this from closely related problems arising in the analysis of network-structured signals, that is, graph signal processing (Shuman et al. 2013), or in estimating network structure from observed multivariate data, that is, structure learning of probabilistic graphical models (Maathuis et al. 2018, Part III).

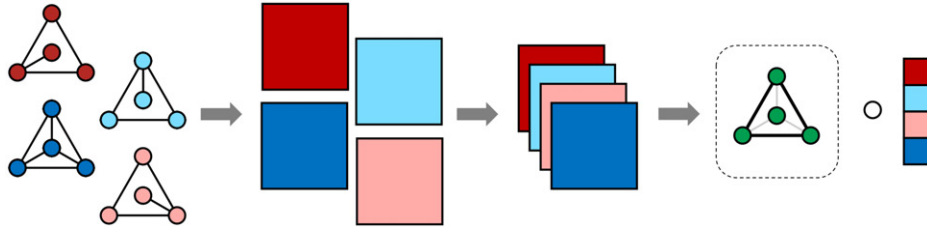


Figure 1. Analysis of Multiple Networks Using Semi-Symmetric Tensor PCA: Given a collection of networks on a common node set (left panel), we express each network using a common matrix representation, for example, adjacency matrix or (regularized) graph Laplacian (second panel), which can then be formed into a semi-symmetric tensor (third panel). The Semi-Symmetric Tensor PCA decomposition (right panel) approximates this tensor as the product of two terms which can be interpreted as a “principal network” on the same node set (green network) and a loading vector across the observations.

1.1. Selected Background and Related Work

The popular “canonical polyadic” (CP) tensor decomposition framework approximates a tensor by a sum of rank-1 components, which are typically estimated using an alternating least squares approach (Kolda and Bader 2009, sec.3). The CP decomposition has been applied as the basis for various forms of tensor PCA, where it is preferred over Tucker-type decompositions because it provides interpretable and ordered components, similar to those resulting from classical (matrix) PCA (Zare et al. 2018). For scalar data arranged in a tensor, CP-based tensor PCA exhibits strong empirical performance and has been fruitfully extended to sparse and high-dimensional settings with efficient algorithms (Allen 2013, 2012), but it is less ideal for network settings. Specifically, CP-based approaches decompose a tensor into rank-1 components, but realistic networks essentially never have exact rank-1 structure, as this would imply a uniform connectivity pattern across all nodes. This problem—developing an efficient tensor decomposition and PCA framework which captures meaningful network structure—motivates the “semi-symmetric tensor PCA” (SST-PCA) decomposition at the heart of this article.

While computational theory for tensor decompositions is well-established, comprehensive *statistical* theory remains an active area of research in the statistics and machine learning communities. Anandkumar, Ge, and Janzamin (2014) give a general analysis of statistical estimation under CP-type models, developing incoherence conditions that allow recovery of multiple CP factors. Extending their analysis to the sparse CP framework of Allen (2012), Sun et al. (2016) establish consistency and sparse recovery under additional sparsity and incoherence assumptions. Consistency results have also been established for tensor decompositions using the Tucker (Zhang and Xia 2018; Zhang and Han 2019) and Tensor-Train (Oseledets 2011; Zhou et al. 2022) frameworks. A recent paper by Han, Willett, and Zhang (2022) considers general estimation problems under a low Tucker-rank assumption and establishes convergence guarantees for a wide range of problems, including low-rank approximation, tensor completion, and tensor regression. Anandkumar et al. (2014) and Janzamin et al. (2019) analyze decompositions of fully-symmetric tensors, such as those arising in certain method-of-moments estimation schemes. The fully-symmetric case imposes additional computational concerns not present in the general case: Kolda (2015) discusses these issues and reviews related work in more detail. The tensor statistical estimation literature is too large to be comprehensively described here and

we refer the reader to the recent survey by Bi et al. (2021) for additional references.

While we are the first to provide rigorous statistical theory for semi-symmetric tensor decompositions, these techniques have already found wide use in the literature. In particular, our SST-PCA builds upon the Network Tensor PCA proposal of Zhang et al. (2019) who apply tensor techniques to brain connectivity in order to find connectomic structures correlated with a variety of interesting behavioral traits. Winter, Zhang, and Dunson (2020) extend this approach to multi-scale network representations, yielding an approach which is robust to the particular brain parcellation used. Despite the empirical successes of these approaches, this work is the first to provide rigorous theoretical guarantees of the type presented in Theorem 1.

Separately, the task of finding common low-rank subspace representations across multiple graphs has been well-studied in the network science community. The recent work of Wang et al. (2021) is particularly closely related to ours, as both approaches employ a tensor decomposition to extract common features among a set of graphs. However, our results significantly generalize and improve upon their findings, as we are able to fit a general “low-rank” model for the principal networks and provide finite-sample guarantees on the quality of both the estimated principal components and the loading vector, whereas that article only provides an asymptotic consistency result for the principal component under a specific rank-1 variant of the Erdős-Rényi model. Similarly, the COSIE proposal of Arroyo et al. (2021) develops a spectral embedding approach for multiple graphs, but does so leveraging techniques from distributed estimation rather than tensor decompositions and under a different probabilistic model than the one considered in this work. Jing et al. (2021) consider a semi-symmetric tensor decomposition approach similar to ours, though they work within the framework of Tucker-type instead of CP-type decompositions. Under a different generative model, but similar initialization and signal-strength assumptions, they achieve the same effective convergence rate as we present in Theorems 1 and 2, suggesting that these rates are indeed optimal for a wide range of network tensor decompositions. Further details and technical comparisons can be found in Section G.1 of the supplementary materials.

The proposed approach is motivated by the statistical analysis of multiple networks, an area of much recent activity. In addition to the embedding analysis described above, we highlight several recent advances in this space including: spectral clustering

of network-valued data (Mukherjee, Sarkar, and Lin 2017); model-based clustering of network valued data (Matias and Miele 2017); probabilistic models and Bayesian estimators for network series (Sewell and Chen 2015; Gollini and Murphy 2016; Durante, Dunson, and Vogelstein 2017); factor models for matrix time series (Chen, Yang, and Zhang 2022); two-sample testing (Ghoshdastidar et al. 2020); and scalar-on-network regression (Arroyo Relión et al. 2019; Guha and Rodriguez 2021; Roy and Michailidis 2022).

1.2. Main Contributions

The contributions of this work are threefold: first, building on a proposal of Zhang et al. (2019), we develop a computationally efficient flexible tensor PCA framework that is able to characterize networks arising from a large class of random graph models; second, we establish statistical consistency of the SST-PCA procedure under fairly weak and general assumptions, centered on a tensor analogue of the low-rank mean model; finally, we establish a novel connection between multilayer network analysis and regularized M -estimation that is of significant independent theoretical interest.

The remainder of this article is organized as follows: Section 2 introduces the “Semi-Symmetric Tensor PCA” decomposition (SST-PCA) and proposes an efficient algorithm for computing the SST-PCA solution. Section 3 establishes statistical properties of SST-PCA via a simple, yet novel, analytical framework; Section 3.1 connects these results to the wider literature on regularized Principal Components Analysis. Section 4 demonstrates the usefulness and flexibility of the SST-PCA framework, both on synthetic data and in a study of the social dynamics of a group of French students. Finally, Section 5 concludes the article and discusses several possible future directions for investigation.

1.3. Notation

Notation used in this article follows that of Kolda and Bader (2009) with two minor modifications: first, in the tensor-matrix product, $\times_k$, we omit a transpose implicitly used by Kolda and Bader: for two matrices $\mathbf{A}, \mathbf{B}$ we have $\mathbf{A} \times_2 \mathbf{B} = \mathbf{AB}$ while they have $\mathbf{A} \times_2 \mathbf{B} = \mathbf{AB}^T$. Consequently, we have $\mathcal{X} \times_k \mathbf{A} \times_k \mathbf{B} = \mathcal{X} \times_k (\mathbf{AB})$ rather than $\mathcal{X} \times_k (\mathbf{BA})$. In general, appropriate axes on which to multiply are clear from context. Second, while $\circ$ denotes the general outer product, we also adopt the convention that, when applied to two matrices of equivalent size, it denotes a matrix product with its transpose, not a higher order product: that is, if $\mathbf{V} \in \mathbb{R}^{p \times k}$, $\mathbf{V} \circ \mathbf{V} = \mathbf{VV}^T$, a $p \times p$ matrix, not a $p \times k \times p \times k$ tensor. Unless otherwise noted, inner products and norms refer to the Frobenius norm and associated inner product ($\|\mathcal{X}\| = \|\text{vec}(\mathcal{X})\|_{\ell_2}$ and $\langle \mathcal{X}, \mathcal{Y} \rangle = \sum_i \text{vec}(\mathcal{X})_i \text{vec}(\mathcal{Y})_i$). We denote the (compact) Stiefel manifold by $\mathcal{V}^{p \times r} = \{\mathbf{V} \in \mathbb{R}^{p \times r} : \mathbf{V}^T \mathbf{V} = \mathbf{I}_{r \times r}\}$. We will make frequent use of the fact that, for any $\mathbf{V} \in \mathcal{V}^{p \times r}$, $\mathbf{V} \circ \mathbf{V}$ is a rank- r matrix of size $p \times p$.

Semi-symmetric tensors are third-order tensors, that is, three dimensional arrays, each slice of which along a fixed axis yields a (real) symmetric matrix. By convention, we take the first two dimensions to be the axes of symmetry, so $\mathcal{X} \in \mathbb{R}^{p \times p \times T}$ is a semi-symmetric tensor if $\mathcal{X}_{..i}$ is a $p \times p$ symmetric matrix for

all $1 \leq i \leq T$. We introduce several notations for nonstandard tensor operations which arise naturally in the semi-symmetric context: for a $p \times p \times T$ semi-symmetric tensor $\mathcal{X}$ and a $p \times k$ matrix $\mathbf{V}$, we define the trace-product $\langle \mathcal{X}; \mathbf{V} \rangle$ as the T -vector whose j th element is given by $\langle \mathcal{X}_{..j}, \mathbf{V} \mathbf{V}^T \rangle = \text{Tr}(\mathbf{V}^T \mathcal{X}_{..j} \mathbf{V})$. For a $p \times p$ symmetric matrix $\mathbf{A}$, $\overline{\text{vec}}\{\mathbf{A}\}$ denotes the $\binom{p}{2}$ -vector formed from the upper triangle (diagonal included) of $\mathbf{A}$; $\overline{\text{vec}}^{-1}\{\cdot\}$ denotes the inverse vector-to-symmetric matrix mapping. We denote the “matricization” of a semi-symmetric tensor $\mathcal{X}$ as $\overline{\mathcal{M}}_3(\mathcal{X}) \in \mathbb{R}^{\binom{p}{2} \times T}$, with each row given by $\overline{\mathcal{M}}_3(\mathcal{X})_{.t} = \overline{\text{vec}}\{\mathcal{X}_{..t}\}$. Note that we use only the upper triangle in the $\overline{\text{vec}}\{\cdot\}$ and $\overline{\mathcal{M}}_3(\cdot)$ operators to avoid redundant columns in the matricized forms of semi-symmetric tensors. Finally, the operator norm $\|\cdot\|_{\text{op}}$ of a symmetric matrix is given by the absolute value of the largest magnitude eigenvalue and the rank- r operator norm of a semi-symmetric tensor is defined as $\|\mathcal{X}\|_{r\text{-op}} = \max_{\mathbf{u}, \mathbf{V} \in \mathbb{B}^T \times \mathcal{V}^{p \times r}} |\langle \mathcal{X}; \mathbf{V}, \mathbf{u} \rangle|$, where $\mathbb{B}^T$ is the unit ball in $\mathbb{R}^T$. The semi-symmetric tensor operator norm is generally difficult to compute, but Proposition 1 in Section 3 gives a tractable upper bound in terms of the operator norm of the individual slices of $\mathcal{X}$.

2. Semi-Symmetric Tensor PCA

Our goal is to develop a tensor PCA framework which is able to capture realistic network structures in its output. While real-world networks have a variety of structures, we focus on networks which are (approximately) low-rank, as these are most well-suited for PCA-type analyses. Many network models are approximately low-rank in expectation, most notably, the random dot-product graph family (Athreya et al. 2018), which includes both latent position and stochastic block models as special cases. The proposed “Semi-Symmetric Tensor PCA” (SST-PCA) factorization approximates a $p \times p \times T$ semi-symmetric tensor $\mathcal{X}$ by a rank- r “principal component” and a loading vector for the final mode, which is typically taken to represent independent replicates observed over time. As seen in the sequel, it is natural in many applications to interpret these as a “principal network” and a “time factor,” but this viewpoint is not essential for the technical developments of this section.

Specifically, the single-factor rank- r SST-PCA approximates $\mathcal{X}$ as

$$\mathcal{X} \approx d \mathbf{V} \circ \mathbf{V} \circ \mathbf{u}$$

where $\mathbf{u} \in \mathbb{R}^T$ is a unit-norm T -vector, $d \in \mathbb{R}_{\geq 0}$ is a scale factor, and $\mathbf{V} \in \mathcal{V}^{p \times r}$ is an orthogonal matrix satisfying $\mathbf{V}^T \mathbf{V} = \mathbf{I}_{r \times r}$. When $r = 1$ and $\mathbf{V}_* = \mathbf{v}_*$ is a unit vector, this coincides with the Network PCA approach of Zhang et al. (2019). This decomposition is related to the standard r -term CP decomposition:

$$\mathcal{X} \approx \sum_{i=1}^r \lambda_i \mathbf{a}_i \circ \mathbf{b}_i \circ \mathbf{c}_i$$

with the additional restrictions $\mathbf{a}_i = \mathbf{b}_i$ and $\mathbf{c}_i = \mathbf{c}_1$ for all $1 \leq i \leq r$ as well as $\langle \mathbf{a}_i, \mathbf{a}_j \rangle = \delta_{ij}$ to ensure orthogonality. These constraints restrict the number of free parameters from $r(2p + T + 1)$ to $pr + T - r(r + 1)/2 + 1$, which significantly improves both

computational performance and estimation accuracy, as will be described in more detail below.

A more general SST-PCA, the $(r_1, \dots, r_K)$ -multi-factor SST-PCA, can be defined by

$$\mathcal{X} \approx \sum_{k=1}^K d_k \mathbf{V}_k \circ \mathbf{V}_k \circ \mathbf{u}_k,$$

where, as before, each $\mathbf{u}_k$ is a unit-norm T -vector, d_k are positive scale factors, and $\mathbf{V}_k \in \mathcal{V}^{p \times r_k}$ are orthogonal matrices used to construct the low-rank principal networks. We emphasize that the $\mathbf{V}_k$ terms need not be of the same dimension, allowing our model to flexibly capture principal components of differing rank. The multi-factor SST-PCA model inherits properties of both the CP and Tucker decompositions: it approximates $\mathcal{X}$ by the sum of multiple simple components like the CP but allows for orthogonal low-rank factors like the Tucker.

Single-factor SST-PCA seeks the best approximation of $\mathcal{X}$ in the tensor Frobenius norm:

$$\arg \min_{\mathbf{u}, \mathbf{V}, d} \|\mathcal{X} - d \mathbf{V} \circ \mathbf{V} \circ \mathbf{u}\|_F^2 \quad \text{subject to } \mathbf{u} \in \mathbb{R}^T, \|\mathbf{u}\| = 1, \\ d \in \mathbb{R}_{\geq 0}, \mathbf{V} \in \mathcal{V}^{p \times r}$$

It is easily seen that solving this problem is equivalent to minimizing the inner product $\langle \mathcal{X}, \mathbf{V} \circ \mathbf{V} \circ \mathbf{u} \rangle = \langle \mathcal{X} \tilde{\times}_3 \mathbf{u}, \mathbf{V} \circ \mathbf{V} \rangle$. While this problem is difficult to optimize jointly in $\mathbf{u}$ and $\mathbf{V}$, closed-form global solutions are available both for $\mathbf{u}$ with $\mathbf{V}$ held constant and for $\mathbf{V}$ with $\mathbf{u}$ held constant. This motivates an alternating minimization (block coordinate descent) strategy comprised of alternating $\mathbf{u}$ and $\mathbf{V}$ updates. Specifically, holding $\mathbf{u}$ constant, the optimal value of $\mathbf{V}$ is simply the leading r eigenvectors of $\mathcal{X} \tilde{\times}_3 \mathbf{u}$ and, holding $\mathbf{V}$ constant, the optimal value of $\mathbf{u}$ is a unit vector in the direction of $[\mathcal{X}; \mathbf{V}]$. Note that, for single factor rank-1 SST-PCA, the $\mathbf{u}$ -update simplifies to $\mathbf{u} \propto \mathcal{X} \tilde{\times}_1 \mathbf{v} \tilde{\times}_2 \mathbf{v}$. Putting these steps together, we obtain the following algorithm for the single-factor rank- r SST-PCA decomposition:

Algorithm 1 Alternating Minimization Algorithm for the Rank- r Single-Factor SST-PCA

- Input: $\mathcal{X}, \mathbf{u}^{(0)}$
 - Initialize: $k = 0$
 - Repeat until convergence:
 - (i) $\mathbf{V}^{(k+1)} = r\text{-eigen}(\mathcal{X} \tilde{\times}_3 \mathbf{u}^{(k)})$
 - (ii) $\mathbf{u}^{(k+1)} = \text{Norm}([\mathcal{X}; \mathbf{V}^{(k+1)}])$
 - (iii) $k := k + 1$
 - Return $\hat{\mathbf{u}} = \mathbf{u}^{(k)}$, $\hat{\mathbf{V}} = \mathbf{V}^{(k)}$, $\hat{d} = r^{-1} \langle \mathcal{X}, \hat{\mathbf{V}} \circ \hat{\mathbf{V}} \circ \hat{\mathbf{u}} \rangle_F$, and $\hat{\mathcal{X}} = \hat{d} \hat{\mathbf{V}} \circ \hat{\mathbf{V}} \circ \hat{\mathbf{u}}$.
-

where $\text{Norm}(\mathbf{x}) = \mathbf{x} / \|\mathbf{x}\|_2$ and $r\text{-eigen}(\mathbf{X})$ denotes the first r eigenvectors of $\mathbf{X}$. When $\mathbf{X}$ has both positive and negative eigenvalues, $r\text{-eigen}(\mathbf{X})$ denotes those eigenvectors whose eigenvalues have the largest magnitudes. Convergence of Algorithm 1 follows from the general block coordinate analysis of Tseng (2001) or from a minor extension of the results of Kofidis and Regalia (2002), noting that the use of an eigendecomposition

allows the $\mathbf{V}$ -update to be solved to global optimality. In Section 3.1, we outline an alternate convergence proof based on recent developments in the theory of regularized (matrix) PCA.

We note here that we consider the case where $\mathbf{V} \circ \mathbf{V}$ is a projection matrix. This simplifies the derivation above and the theoretical analysis presented in the next section, but is not necessary for the approach. In situations where the eigenvalues of $\mathbf{V} \circ \mathbf{V}$ are allowed to vary, the $\mathbf{V}$ -update in Algorithm 1 can be replaced by $\mathbf{V}^{(k+1)} = \mathbf{D}^{1/2} r\text{-eigen}(\mathcal{X} \tilde{\times}_3 \mathbf{u}^{(k)})$ with $\mathbf{D}$ being the matrix containing the leading r eigenvalues of $\mathcal{X} \tilde{\times}_3 \mathbf{u}^{(k)}$ on the diagonal. Standard eigensolvers return elements of $\mathbf{D}$ alongside the eigenvectors, so this approach does not change the complexity of our approach. For the theoretical analysis in Section 3, d can be taken to be the r th eigenvalue of the true network factor, $\mathbf{V}_* \circ \mathbf{V}_*$, with minor modification.

2.1. Deflation and Orthogonality

While Algorithm 1 provides an efficient approach for estimating a single SST-PCA factor, the multi-factor case is significantly more challenging. The difficulty of the multi-factor case is a general characteristic of CP-type decompositions, posing significant computational and theoretical challenges, discussed at length by Han et al. (2024) among others. To avoid these difficulties, we adapt a standard greedy (deflation) algorithm to the SST-PCA setting, based on repeatedly deflating $\mathcal{X} \leftarrow \mathcal{X} - \hat{d} \hat{\mathbf{u}} \circ \hat{\mathbf{V}} \circ \hat{\mathbf{V}}$ and applying Algorithm 1; this sequential deflation approach to CP-type decompositions builds on well-known properties of the matrix power method and has been adapted to tensors independently by several authors, including Kolda, Bader, and Kenny (2005) and Allen (2012). While this approach was originally applied heuristically, Mu, Hsu, and Goldfarb (2015) and Ge et al. (2021) have shown conditions under which the deflation approach is able to recover an optimal solution. As Mackey (2008) notes, classical (Hotelling) deflation fails to provide orthogonality guarantees when approximate eigenvectors are used, such as those arising from regularized variants of PCA. To address this, he proposes several additional deflation schemes with superior statistical properties. We discuss tensor versions of these schemes in Section A.1 of the supplementary materials.

We note that Algorithm 1 is only guaranteed to reduce the norm of the residuals and may not reduce the tensor rank at each iteration: in fact, in some circumstances, the tensor rank may actually be increased by this approach. We believe this is not a weakness of the approach: as signal is removed from a data tensor, the remaining residuals should increasingly resemble pure noise, which has high tensor rank almost surely (Kolda and Bader 2009, sec. 3.1). Similarly, as we remove estimated signal, the unexplained residual variance should decrease. While classical PCA does reduce the rank of the residual matrix at each iteration, this is an attractive, but incidental, property of linear algebra and not an essential statistical characteristic.

While iterative deflation approaches to multi-factor SST-PCA are simple to implement and provide robust orthogonality guarantees, in some circumstances it may be preferable to modify Algorithm 1 to estimate multiple tensor PCs simultaneously. We provide additional details of the simultaneous approach in Section A.2 of the supplementary materials along with a discussion

of its theoretical properties. In practice, we have found deflation approaches to be more robust to initialization and rank selection and we recommend them in practice.

2.2. Algorithmic Considerations

The computational cost of [Algorithm 1](#) is dominated by the eigendecomposition appearing in the $\mathbf{V}$ -update. Standard algorithms scale with complexity $\mathcal{O}(p^3)$, which can be reduced to $\mathcal{O}(p^2 r)$ since we only require r leading eigenvectors. Computation of the trace product $[\mathcal{X}; \mathbf{V}]$ is similarly expensive with complexity $\mathcal{O}(p^2 T r^2)$ but can be trivially parallelized across slices of $\mathcal{X}$. Taken together, these give an overall per iteration complexity of $\mathcal{O}(p^2 \max\{T r^2/C, r\})$ when C processing units are used. In practice, we have found the cost of the eigendecomposition to dominate in our experiments. As our theory below suggests, only a relatively small number of iterations, I , are required to achieve statistical convergence with I decaying rapidly in the signal-to-noise ratio of the problem.

[Algorithm 1](#) given above closely parallels the classical power method for computing singular value decompositions. As such, many techniques from that literature can be used to improve the performance of our approach or to adapt it to additional computational constraints. While a full review of this literature is beyond the scope of this article, we highlight the use of sketching techniques for particularly large data, one-pass techniques for streaming data, and distributed techniques (Tropp et al. 2017, 2019; Li et al. 2021). When applied in the context of networks, it is particularly advantageous to take advantage of sparsity in $\mathcal{X}$ in both the eigendecomposition and tensor algebra steps (Phipps and Kolda 2019).

3. Consistency of SST-PCA

We analyze SST-PCA under a tensor analogue of the spiked covariance model popularized by Johnstone (2001) or the low-rank mean model of Hoff (2007) in the matrix context. Specifically, we consider data generated with signal corresponding to a rank- k principal component $\mathbf{V}_* \circ \mathbf{V}_* = \mathbf{V}_* \mathbf{V}_*^T$ for some fixed $\mathbf{V}_* \in \mathcal{V}^{p \times k}$ and a loading vector $\mathbf{u}_* \in \mathbb{B}_T$ and noise drawn from a semi-symmetric sub-Gaussian tensor, $\mathcal{E}$:

$$\mathcal{X} = d \mathbf{V}_* \circ \mathbf{V}_* \circ \mathbf{u}_* + \mathcal{E}, \quad (1)$$

wherein $d \in \mathbb{R}_{>0}$ is a measure of signal strength, roughly analogous to the square root of the leading eigenvalue of the spiked covariance model. (Montanari and Richard (2014) consider a similar construction, albeit with stronger symmetry restrictions.) In this setting, the following consistency result holds:

Theorem 1. Suppose $\mathcal{X}$ is generated from the semi-symmetric model in (1) with elements of $\mathcal{E}$ each being independently mean 0, σ^2 -sub-Gaussian, subject to symmetry constraints. Suppose further that the initialization $\mathbf{u}^{(0)}$ satisfies

$$|1 - \langle \mathbf{u}^{(0)}, \mathbf{u}_* \rangle| \leq \tan^{-1}(0.5) - \frac{5 \|\mathcal{E}\|_{r\text{-op}}}{d(1-c)}$$

for some arbitrary $c < 1$. Finally, assume $d \gtrsim r\sqrt{T}\sigma(\sqrt{p} + \sqrt{\log T})$. Then, the output of [Algorithm 1](#) applied to $\mathcal{X}$

satisfies

$$\min_{\epsilon=\pm 1} \|\epsilon \mathbf{u}_* - \hat{\mathbf{u}}\|/\sqrt{T} \lesssim \frac{\sigma r \sqrt{p}}{d(1-c)}.$$

with high probability. $\lesssim$ indicates an inequality holding up to a universal constant factor and $\lesssim$ denotes an inequality holding up to a constant factor and a term scaling as $\sqrt{\log T}$.

A similar result holds for the $\mathbf{V}$ -factor:

Theorem 2. Under the same conditions as [Theorem 1](#), the output of [Algorithm 1](#) applied to $\mathcal{X}$ satisfies, with high probability,

$$\min_{\mathbf{O} \in \mathcal{V}^{r \times r}} \frac{\|\mathbf{V}_* - \hat{\mathbf{V}}\mathbf{O}\|_F}{\sqrt{pr}} \lesssim \frac{\sigma r \sqrt{T}}{d(1-c)}.$$

In both results, the ϵ and $\mathbf{O}$ terms reflect the fact that, like classical PCA, the SST-PCA factors $\mathbf{u}_*$ and $\mathbf{V}_*$ are defined only up to arbitrary rotation.

We highlight that these results are first-order comparable to those for classical PCA attained by Nadler (2008), though with more strenuous conditions on the initialization. (The results of the matrix power method almost surely do not depend on the choice of initialization, but this property is unique to the classical (unregularized) eigenproblem and does not hold in general.) A full proof of [Theorems 1](#) and [2](#) is given in the supplementary materials, but we outline the key aspects of our proof strategy in [Section 3.2](#). A highlight of our approach is that it relies only on the Davis-Kahan theorem and standard concentration inequalities, while Nadler uses sophisticated matrix perturbation results. The fact that comparable results are obtained for the matrix and semi-symmetric tensor cases suggests that our results are essentially the best that can be obtained for this problem using elementary non-asymptotic techniques.

The initialization condition of [Theorem 1](#) essentially assumes initialization within $2 \tan^{-1}(0.5) \approx 53.1^\circ$ of the true $\mathbf{u}_*$ factor, with additional accuracy needed to deal with higher-noise scenarios. In low-dimensions, this condition is easily satisfied, but it becomes more strenuous in the large T setting. We believe this condition to be, in part, an artifact of our proof technique: our experiments in [Section 4](#) suggest that initialization in the correct orthant is typically sufficient. (Zhang and Xia (2018) show that some sort of initialization condition is in fact unavoidable.) This is particularly compelling in the network setting, where it is reasonable to assume that all elements of $\mathbf{u}_*$ are positive, that is, that the network does not swap a large number of edges simultaneously, flipping the loading on the principal network. In settings where $\mathbf{u}_*$ can not be assumed to be elementwise positive, [Algorithm 1](#) is sufficiently computationally efficient to allow for repeated random initializations.

We note that our results are derived for root mean squared error under sub-Gaussian noise for analytical simplicity, but that our proof technique could be applied more broadly. In particular, if one assumes IID Bernoulli noise, corresponding to randomly corrupted edge indicators in the network setting, the results of Vu (2011) lead to significantly tighter bounds. In certain high-dimensional problems, elementwise error bounds on $\mathbf{u}$ and $\mathbf{V}$ may be more useful than an overall Root Mean Squared Error (RMSE) type bound and could be derived using the ℓ_∞ eigenvector perturbation results of Fan, Wang, and

Zhong (2018) or Damle and Sun (2020). Finally, we note that because our analysis leverages Davis-Kahan (uniform) bounds, we do not assume any cancellation among elements of $\mathcal{E}$, allowing our proofs to be applied in a dynamic adversarial setting, so long as $\|\mathcal{E}\|_{r\text{-op}}$ remains bounded at each iteration. Replacing the Davis-Kahan bounds with high-probability bounds such as those of O’Rourke, Vu, and Wang (2018) could lead to tighter bounds or less stringent initialization and signal conditions. Theorem D.4 in the supplementary materials extends Theorems 1 and 2 to the $K > 1$ setting.

3.1. Connection with Regularized PCA

At this point, the reader may wonder about the connection between traditional PCA and SST-PCA: after all, our approach endows the space of semi-symmetric tensors with a real inner product, treating it as essentially Euclidean, and achieves the same estimation error as classical PCA under the appropriate rank-one model. As discussed next, there is an equivalence between SST-PCA and a certain form of regularized matrix PCA not previously considered in the literature. The regularization implicit in this equivalence is key to understanding how the proposed approach can perform well on large graphs and avoid the pitfalls of standard PCA in high-dimensional settings (Johnstone and Lu 2009).

In order to express a $p \times p \times T$ -dimensional semi-symmetric tensor as a matrix, it is natural to form a $T \times \binom{p}{2}$ data matrix by vectorizing one triangle of each tensor slice and combining the vectorized slices as rows of a matrix. For a given semi-symmetric tensor, $\mathcal{X}$, let us denote the corresponding data matrix as $\mathbf{X} = \overline{\mathcal{M}}_3(\mathcal{X})$ where $\overline{\mathcal{M}}_k(\cdot)$ denotes tensor matricization of the upper triangle of each slice preserving the k th mode, that is $\mathbf{X}_{i\cdot} = \text{vec}\{\mathcal{X}_{\cdot\cdot i}\}$. Performing classical PCA on this matrix will have estimation error proportional to $\sigma \sqrt{\binom{p}{2}}/d \approx \sigma p/d$, which can be significantly worse than the $\sigma r \sqrt{p}/d$ rate attained by SST-PCA for graphs having rank $r \ll \sqrt{p}$. Additionally, classical PCA imposes no requirements on the estimated principal component and does not imply that the principal network given by $\text{vec}^{-1}\{\hat{\mathbf{v}}\}$ has a low-rank structure (Table 1).

Low-rank structure of the principal network can be reimposed by adding an additional constraint on the right-singular vector: that is, by solving the regularized singular value problem

$$\arg \max_{\mathbf{u} \in \mathbb{B}^T, \mathbf{v} \in \mathbb{B}^{\binom{p}{2}}} \mathbf{u}^T \overline{\mathcal{M}}_3(\mathcal{X}) \mathbf{v} \quad \text{subject to } \text{rank}(\text{vec}^{-1}\{\mathbf{v}\}) = k.$$

Finally, note that standard reformulation of the singular value problem as an eigenproblem via Hermitian dilation yields an equivalent regularized eigenvalue problem:

$$\arg \max_{\mathbf{x} = (\mathbf{u}, \mathbf{v}) \in \mathbb{B}^{T + \binom{p}{2}}} \mathbf{x}^T \begin{pmatrix} \overline{\mathcal{M}}_3(\mathcal{X})^T \overline{\mathcal{M}}_3(\mathcal{X}) & \mathbf{0} \\ \mathbf{0} & \overline{\mathcal{M}}_3(\mathcal{X}) \overline{\mathcal{M}}_3(\mathcal{X})^T \end{pmatrix} \mathbf{x} \\ \text{subject to } \text{rank}(\text{vec}^{-1}\{\mathbf{x}_{-(1:T)}\}) = k$$

While these formulations are much more unwieldy than the tensor formulation, they reveal connections between our approach and the rich literature on sparse PCA. Specifically, our SST-PCA Algorithm 1 can be interpreted as a variant of the truncated

Table 1. Comparison of SST-PCA with matrix-based approaches.

Method	Data dimension	Aligned RMSE	
		$\mathbf{u}$ -Factor	$\mathbf{v}/\mathbf{V}$ -Factor
Classical PCA	$T \times p$	$\frac{\sigma \sqrt{p}}{d}$	$\frac{\sigma \sqrt{T}}{d}$
Matricization + Classical PCA	$T \times \binom{p}{2}$	$\frac{\sigma p}{d}$	$\frac{\sigma \sqrt{T}}{d}$
SST-PCA (rank 1)	$p \times p \times T$	$\frac{\sigma \sqrt{p}}{d}$	$\frac{\sigma \sqrt{T}}{d}$
SST-PCA (rank r)	$p \times p \times T$	$\frac{\sigma r \sqrt{p}}{d}$	$\frac{\sigma r \sqrt{T}}{d}$

NOTE: Under a “signal plus noise” model with signal strength d and σ^2 -sub-Gaussian noise, rank-1 SST-PCA achieves the same accuracy as classical PCA on a $T \times p$ matrix. For the network PCA problem, the naïve approach of representing each observed network as a row vector and performing classical (matrix) PCA performs worse than SST-PCA by a factor of $\sqrt{p}$. For large graphs well approximated by low-rank graphs, the improvements attained by SST-PCA are particularly meaningful. Aligned RMSE refers to the scaled and rotated RMSE appearing in Theorems 1 and 2, specifically $\text{ARMSE}(\mathbf{V}, \hat{\mathbf{V}}) = \min_{\mathbf{O} \in \mathbb{R}^{r \times r}} \|\mathbf{V} - \hat{\mathbf{V}} \mathbf{O}\|_F / \sqrt{pr}$ for two matrices $\mathbf{V}, \hat{\mathbf{V}} \in \mathbb{R}^{p \times r}$. For vectors, for example, the $\mathbf{u}$ -factor, aligned RMSE reduces to comparing $\mathbf{u}$ against both $\hat{\mathbf{u}}$ and $-\hat{\mathbf{u}}$. Derivation of the error rates for classical PCA under a “signal plus noise” model appear in Section C of the supplementary materials.

power method considered by Yuan and Zhang (2013) and Ma (2013), with the truncation to a k -sparse vector being replaced by truncation to a matrix with only k nonzero eigenvalues. In this form, it is clear that Yuan and Zhang’s (2013) convergence analysis of the Truncated Power Method can be applied to give an alternate proof of the convergence of Algorithm 1.

While we do not consider minimax optimality in this article, we note that substantially similar approaches have been shown to be minimax optimal under comparable assumptions. In particular, Birnbaum et al. (2013) show that classical PCA is rate-optimal under a dense signal model and that Ma’s (2013) version of the truncated power method is optimal under a sparse signal model. Relatedly, Donoho and Gavish (2014) establish optimality of a soft-thresholded SVD under the matrix version of the “signal plus noise” model. Because our method attains comparable convergence rates, we conjecture that SST-PCA is minimax rate-optimal up to a logarithmic factor under conditions substantially similar to those of Theorem 1.

This “rank-unvec” constraint appearing above has not previously appeared in the literature, but arises naturally from the low-rank structure random dot-product graphs. A similar constraint is implicit in certain unrelaxed formulations of the linear matrix sensing problem, but typically not analyzed as such (Candès and Plan 2011). Despite the non-convexity of this constraint, it is both computationally and theoretically tractable due to the nice properties of low-rank projections. We believe that this type of constraint can be applied more broadly in the analysis of network data, for example, a variant of the network classification scheme considered by Arroyo Relión et al. (2019) with the coefficient matrix representing a low-rank graph rather than a sparse set of edges.

3.2. Sketch of Proof of Theorems 1 and 2

The proofs of Theorems 1 and 2 are outlined next, with full proofs provided in Section D of the supplementary materials. Analytical difficulties arise from (i) the non-convexity of tensor decompositions and lack of globally convergent algorithms; and (ii) the lack of Davis-Kahan type results for tensor factoriza-

tions. To address these, we first demonstrate how matrix Davis-Kahan results can be extended to the tensor case using the regularized PCA reduction discussed in the previous section; we then perform an intricate convergence analysis of [Algorithm 1](#) to show that SST-PCA estimates converge to a small neighborhood of the true parameter values. Hardt and Price's (2014) analysis of the matrix power method provides useful background.

We first establish the following tail bound on the size of the noise tensor $\mathcal{E}$ in terms of its semi-symmetric rank- r operator norm:

Proposition 1. The semi-symmetric operator norm, $\|\mathcal{E}\|_{r\text{-op}} = \max_{\mathbf{u}, \mathbf{V} \in \mathbb{B}^T \times \mathcal{V}^{p \times r}} \|[\mathcal{E}; \mathbf{V}]^T \mathbf{u}\|$ can be deterministically bounded above by $r\sqrt{T} \max_i \lambda_{\max}(\mathcal{E}_{..i})$. Furthermore, if the elements of $\mathcal{E}$ are independently σ^2 -sub-Gaussian, subject to symmetry constraints, we have

$$\|\mathcal{E}\|_{r\text{-op}} \leq cr\sqrt{T}\sigma \left(\sqrt{p} + \sqrt{\log T} + \delta \right)$$

with probability at least $1 - 4e^{-\delta^2}$, for some absolute constant c .

The deterministic claim follows by bounding the elements of $[\mathcal{E}; \mathbf{V}]$ separately as $[\mathcal{E}; \mathbf{V}]_i = \sum_{j=1}^r \lambda_j(\mathcal{E}_{..i}) \leq r\lambda_{\max}(\mathcal{E}_{..i})$ and applying the standard result that $\|\mathbf{x}\|_2 \leq \sqrt{T}\|\mathbf{x}\|_\infty$ for any $\mathbf{x} \in \mathbb{R}^T$. The probabilistic claim follows from standard bounds on the maximum eigenvalue of sub-Gaussian matrices and a union bound (Vershynin 2018, Theorem 4.4.5). Note that, in the common case where the edges of $\mathcal{X}$ are binary, $\mathcal{E}$ can be taken to be a matrix of $\{-1, 0, 1\}$ values: in this case, elements of $\mathcal{E}$ are independently sub-Gaussian with $\sigma^2 = 1$, so the σ factors in [Proposition 1](#) and [Theorems 1](#) and [2](#) vanish.

We combine the above result with a deterministic bound on the accuracy of [Algorithm 1](#):

Proposition 2. Under the conditions of [Theorem 1](#) with $\mathcal{E}$ held fixed, the result of [Algorithm 1](#) applied to $\mathcal{X}$ satisfies the following:

$$\min_{\epsilon = \pm 1} \|\epsilon \mathbf{u}^* - \hat{\mathbf{u}}\|_2 \leq \frac{8\sqrt{2} \|\mathcal{E}\|_{r\text{-op}}}{d(1-c)}$$

so long as $\|\mathcal{E}\|_{r\text{-op}} < d$ and $\mathbf{u}^{(0)}$ satisfies the following initialization condition, for some arbitrary $c < 1$:

$$|1 - \langle \mathbf{u}^{(0)}, \mathbf{u}_* \rangle| \leq \tan^{-1}(0.5) - \frac{5 \|\mathcal{E}\|_{r\text{-op}}}{d(1-c)}.$$

Combining [Propositions 1](#) and [2](#) yields [Theorem 1](#). [Theorem 2](#) follows by a similar argument. We establish [Proposition 2](#) using an iterative analysis of [Algorithm 1](#) which controls the error in $\mathbf{u}^{(k)}$ in terms of $\mathbf{u}^{(k-1)}$ and repeats this to convergence. Applying Yu, Wang, and Samworth's (2015) variant of the Davis-Kahan theorem to the $\mathbf{V}$ -update, we find

$$\begin{aligned} & \|\sin \Theta(\mathbf{V}^*, \mathbf{V}^{(k+1)})\|_F \\ & \leq 2\sqrt{r}|1 - \cos \angle(\mathbf{u}_*, \mathbf{u}^{(k)})| + 2\sqrt{r} \|\mathcal{E}\|_{r\text{-op}} d. \end{aligned}$$

Similarly, applying the Davis-Kahan theorem to the $\mathbf{u}$ -update yields

$$\begin{aligned} |\sin \angle(\mathbf{u}_*, \mathbf{u}^{(k+1)})| & \leq 2|1 - \langle \mathbf{V}_*, \mathbf{V}^{(k+1)} \rangle|^4 \\ & \quad + 4r \frac{\|\mathcal{E}\|_{r\text{-op}}}{d} + 2r^2 \frac{\|\mathcal{E}\|_{r\text{-op}}^2}{d^2} \end{aligned}$$

Due to the nonlinear $\text{Norm}(\cdot)$ function, this is not as simple as the $\mathbf{V}$ -update and requires applying the Davis-Kahan theorem to the matrix pair $\tilde{\mathbf{u}}^{(k+1)} \circ \tilde{\mathbf{u}}^{(k+1)}$ and $d^2 \mathbf{u}_* \circ \mathbf{u}_*$, where $\tilde{\mathbf{u}}^{(k+1)} = [\mathcal{X}; \mathbf{V}^{(k+1)}]$ is the pre-normalized iterate. Because $\mathbf{u}^{(k+1)}$ is an eigenvector of $\tilde{\mathbf{u}}^{(k+1)} \circ \tilde{\mathbf{u}}^{(k+1)}$ by construction, the eigenvector bound provided by Davis-Kahan is exactly what is needed to control $\theta_{\mathbf{u}_{k+1}} = \angle(\mathbf{u}_*, \mathbf{u}^{(k+1)})$.

With these two bounds in hand, we can bound the sine of the $\mathbf{V}$ -angle in terms of the cosine of the $\mathbf{u}$ -angle and vice versa. To connect these, it suffices to assume we are in the range of angles such that $|\sin \theta| \geq 2|1 - \cos \theta|$ or $\theta \in [0, 2 \tan^{-1}(0.5)) \approx [0, 53.1^\circ)$. In order to ensure a substantial contraction at each step, we make the slightly stronger assumption that

$$2|1 - \cos \theta_{\mathbf{u}_k}| \leq c_u |\sin \theta_{\mathbf{u}_k}| \quad \text{and} \quad 2|1 - \cos \theta_{\mathbf{v}_k}| \leq c_v |\sin \theta_{\mathbf{v}_k}|$$

for all k and for some fixed $c_u, c_v < 1$. From here, we iterate our $\mathbf{u}$ - and $\mathbf{V}$ -update bounds to show that the error in the $\mathbf{u}$ -iterates is bounded above by

$$|\sin \theta_{\mathbf{u}_k}| \leq c^k |\sin \theta_{\mathbf{u}^{(0)}}| + \frac{8 \|\mathcal{E}\|_{r\text{-op}}}{d} \sum_{i=0}^{k-1} c^i$$

where $c = c_u c_v$. Taking the $k \rightarrow \infty$ limit, this gives the error bound $\frac{8 \|\mathcal{E}\|_{r\text{-op}}}{d(1-c)}$ for the final estimate $\hat{\mathbf{u}} = \lim_{k \rightarrow \infty} \mathbf{u}^{(k)}$. Finally, we note that given the geometric series structure of the above bound, only a logarithmically small number of iterations are required for the iterates to converge up to the noise bound of the data.

An important technical element of our proof is to ensure that the iterates remain within the strict (c_u, c_v) -contraction region for all k . While this follows immediately for the noiseless case, the noisy case is more subtle, requiring us to balance the non-expansive error from our initialization with the effect of the noise $\mathcal{E}$ which recurs at each iteration. To ensure this, we need our iterates to be bounded away from the boundary of the contraction region so that the sequence of “contract + add noise” does not increase the total error at any iterate. We term this non-expansive region the “stable interior” of the contraction region. Simple algebra shows that assuming $\theta_{\mathbf{u}_k}$ is in the stable interior, that is,

$$|1 - \cos \theta_{\mathbf{u}_k}| \leq \tan^{-1}(0.5) - \frac{5 \|\mathcal{E}\|_{r\text{-op}}}{d(1 - c_u c_v)},$$

implies that $\theta_{\mathbf{u}_{k+1}}$ is in the stable interior so it suffices to make this stronger assumption on the initialization $\theta_{\mathbf{u}^{(0)}}$ only. This style of algorithm-structured analysis has recently found use in the analysis of non-convex problems: notable examples include recent work by Fan et al. (2018), Sun et al. (2016), Wang, Lu, and Liu (2014), and Zhao, Liu, and Zhang (2018). In particular, the contraction assumptions above parallel the *Restricted Correlated Gradient* condition of Han, Willett, and Zhang's (2022) low Tucker-rank framework. Compared with those papers, our result holds under much more general assumptions, only requiring bounds on the signal-to-noise ratio of the problem and a loose bound on initialization. Specifically, we do not require standard high-dimensional assumptions of sparsity, restricted strong convexity, or incoherence.

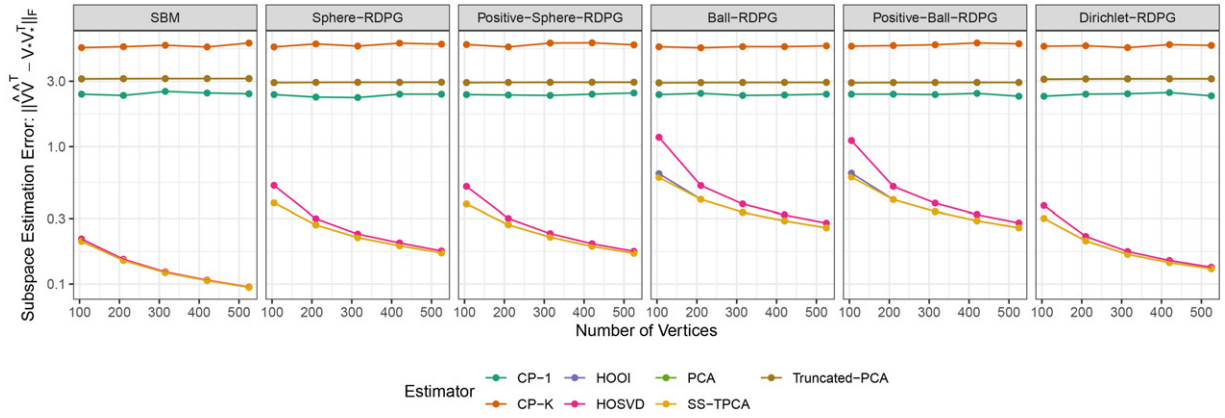


Figure 2. Comparison of low-rank subspace estimation under different network PCA models, under $T = 20$ iid samples from rank-5 graphs of varying size. SST-PCA (gold) outperforms competitor methods; the HOSVD (pink) and HOOI (purple) Tucker decompositions also estimate V_* well but lack the interpretability of SST-PCA due to the non-diagonal Tucker core. Matricization-based approaches, both with and without an additional truncation step, and CP-ALS approaches are not competitive. SBM samples are generated with intra-group connection probability $p = 0.8$ and inter-group connection probability $q = 0.2$. Dirichlet-RDPG refers to latent positions sampled from the five-dimensional simplex using a Dirichlet distribution with parameter $\alpha = 0.3 * \mathbf{1}_5$. Samples are generated and tensor decompositions are performed using the *igraph* and *rTensor* packages, respectively.

4. Empirical Results

4.1. Simulation Studies

We demonstrate the effectiveness of SST-PCA under various conditions and give empirical evidence to support the theoretical claims of the previous two sections. Specifically, we seek to demonstrate the following four claims empirically: (i) SST-PCA outperforms classical PCA methods and generic tensor decompositions in the analysis of network valued data; (ii) SST-PCA exhibits rapid *computational* convergence to a region surrounding the true parameter value; (iii) SST-PCA exhibits classical *statistical* convergence rates as T and p vary; and (iv) SST-PCA is robust to the choice of initialization, $\mathbf{u}^{(0)}$.

Simulation I - Methods Comparison. We generate $\mathcal{X}$ using IID samples from different variants of the Random Dot Product Graph framework, including a stochastic block model and latent position graphs on the simplex, unit sphere, and unit ball. In each case, we simulate $T = 20$ graphs with a rank-5 structure, but vary the number of nodes from 105 to 525. We compare SST-PCA with several potential competitors, including classical (vectorized) PCA, classical PCA followed by a low-rank truncation step, a rank-1 CP decomposition, a rank-5 CP decomposition, and a Tucker decomposition of order (5, 5, 1). The CP decompositions are estimated using alternating least squares, while we consider both the Higher-Order SVD (HOSVD) and the Higher-Order Orthogonal Iterations (HOOI) approaches to estimating Tucker decompositions; see Kolda and Bader (2009) for additional background. The results are depicted in Figure 2, where we report the subspace estimation error for each method. (We do not report estimation error for the $\mathbf{u}$ -factor as it is uninteresting for IID samples, but all methods estimate both terms.) SST-PCA consistently outperforms naïve PCA and other CP approaches. The HOSVD and HOOI Tucker-type approaches are known to be approximately optimal denoisers under this model, but SST-PCA achieves slightly superior performance in all scenarios considered, while retaining superior interpretability.

Simulation II - Computational and Statistical Convergence.

We generate $\mathcal{X}$ from the model from the model $\mathcal{X} = d V_* \circ V_* \circ \mathbf{u}_* + \mathcal{E}$, with $T = 20$ observations and $p = 200$ vertices. Because our main theorems do not place any restrictions on V_* , we draw V_* uniformly at random from the rank- r Stiefel manifold. We fix $\mathbf{u}_* = \mathbf{1}_T / \sqrt{T}$ and initialize Algorithm 1 with $\mathbf{u}^{(0)}$ drawn randomly from the set of positive unit vectors. Finally, we generate $\mathcal{E}$ such that each slice of $\mathcal{E}$ is an independent draw from the Gaussian orthogonal ensemble (GOE): that is, the off-diagonal elements are drawn from independent standard normal distributions and the diagonal elements are independent samples from a $\mathcal{N}(0, 2)$ distribution. We vary the signal strength $d = 15r^{-1/4}$ to preserve a signal to noise ratio of slightly more than one (note $\sqrt{p} \approx 14.14$). Figure 3 depicts the convergence of the iterates of Algorithm 1 under this regime. It is clear that the iterates of Algorithm 1 achieve *statistical convergence*, in the sense of nondecreasing estimation error, in approximately one tenth the iterations needed to achieve computational convergence.

Simulation III - Convergence Rates and Initialization Robustness.

We generate $\mathcal{X}$ as before, but fix $r = 1$ so V_* is drawn uniformly at random from the rank-1 Stiefel manifold, that is, the unit sphere. We now also let $\mathbf{u}_*$ be generated randomly under two different mechanisms: $\mathbf{u}_*$ drawn uniformly at random from the unit sphere and $\mathbf{u}_*$ drawn uniformly at random from the portion of the unit sphere in the first (positive) orthant, that is, nonnegative unit vectors. As before, each slice of $\mathcal{E}$ is independently drawn from the GOE. From well-known bounds on the GOE, we have $\|\mathcal{E}\|_{\text{op}} \approx \sqrt{p \log T}$, giving an effective signal-to-noise ratio of $d / \sqrt{p \log T}$ for all experiments. We consider three approaches to initialize Algorithm 1: (i) oracle initialization - $\mathbf{u}^{(0)} = \mathbf{u}_*$; (ii) random initialization where $\mathbf{u}^{(0)}$ is drawn at random from the unit sphere; and (iii) “stable” initialization with $\mathbf{u}^{(0)} = \mathbf{1}_T / \sqrt{T}$.

Figure 4 shows the results of our study in the case of fixed T (number of samples) and varying p (network size) from 10 to 110. We present three measures of accuracy: (i) the normalized reconstruction error between $\hat{\mathcal{X}} = \hat{d} \hat{V} \circ \hat{V} \circ \hat{\mathbf{u}}$ and the

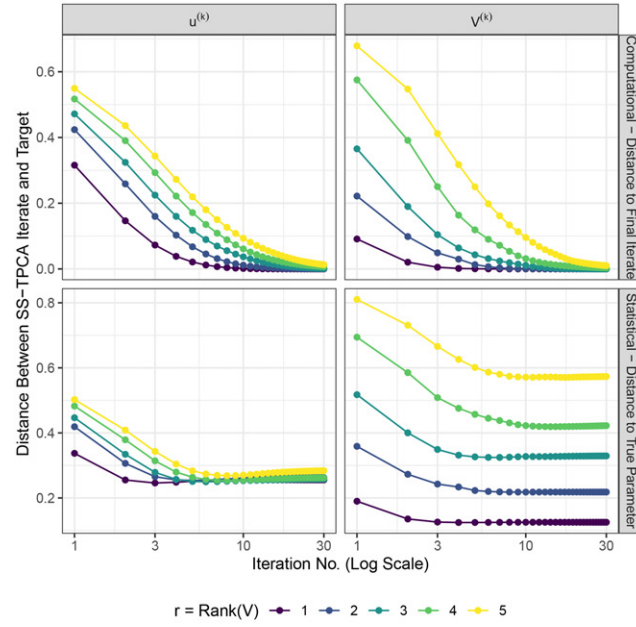


Figure 3. Comparison of the *computational* and *statistical* convergence of Algorithm 1. While SST-PCA takes approximately 30 iterations to converge to $(\hat{\mathbf{u}}, \hat{\mathbf{V}})$, the intermediate iterates $(\mathbf{u}^{(k)}, \mathbf{V}^{(k)})$ have essentially the same statistical accuracy after $k = 4$ iterates.

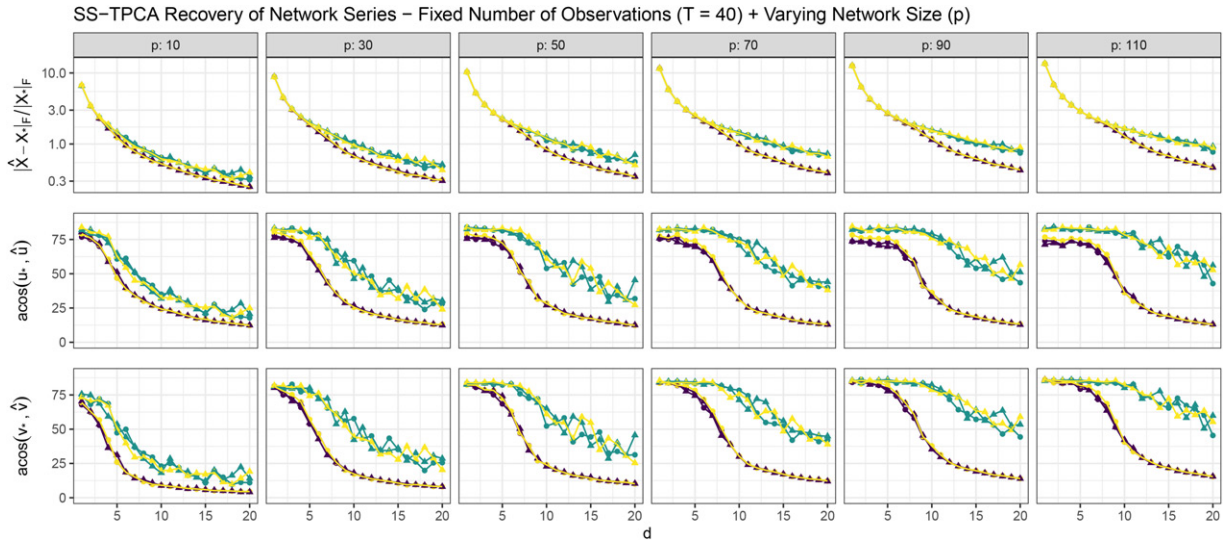


Figure 4. Performance of Algorithm 1 for recovery of a rank-1 semi-symmetric tensor of norm d with added noise from a Gaussian Orthogonal Ensemble. Tensor recovery is accurate across several measures including the normalized reconstruction error between $\hat{\mathcal{X}}$ from Algorithm 1 and the signal tensor $\mathcal{X}_* = d \mathbf{V}_* \circ \mathbf{V}_* \circ \mathbf{u}_*$ (top row), the angle between $\mathbf{u}_*$ and $\hat{\mathbf{u}}$ (middle row), and the angle between $\mathbf{V}_*$ and $\hat{\mathbf{V}}$ (bottom row). As expected, oracle initialization (purple, $\mathbf{u}_0 = \mathbf{u}_*$) consistently performs the best of the initialization schemes considered, but random initialization (green) and “stable” initialization (yellow, to the all-ones vector) also perform well. In scenarios where $\mathbf{u}_*$ is restricted to have all positive elements, stable initialization performs almost indistinguishably from oracle initialization; in scenarios where $\mathbf{u}_*$ is allowed to have both positive and negative elements, stable initialization performs essentially as well as random initialization, suggesting that our method is quite robust to the specific choice of initialization. We fix $T = 40$ observations and vary the dimension of $\mathbf{V}_*$ from 10 (far left column) to 110 (far right column); as would be expected, performance does decay for larger dimensions, but the decay is rather slow, consistent with our theoretical results.

signal tensor $\mathcal{X}_* = d \mathbf{V}_* \circ \mathbf{V}_* \circ \mathbf{u}_*$; (ii) the angle between $\mathbf{u}_*$ and $\hat{\mathbf{u}}$; and (iii) the angle between the subspaces spanned by $\mathbf{V}_*$ and $\hat{\mathbf{V}}$, calculated as $\cos^{-1} \left[\sigma_{\min}(\mathbf{V}_*^T \hat{\mathbf{V}}) \right]$, where $\sigma_{\min}(\cdot)$ denotes the last (nonzero) singular value. Note that, for very accurate recovery, $\cos^{-1}(\cdot)$ introduces nonlinear artifacts near zero: the reconstruction error of $\hat{\mathcal{X}}$ more clearly exhibits the expected parametric rate, which can also be seen in Figure 2. Consistent with Theorems 1 and 2, we observe errors that decay rapidly in d for all measures and scenarios. Comparing $\mathbf{u}_*$ drawn from the positive orthant or the entire unit sphere, we observe

that random initialization does well in both cases, with accuracy only slightly worse than oracle initialization, and that stable initialization achieves essentially the same performance as oracle initialization for positive $\mathbf{u}_*$.

Figure 5 presents analogous results taking the network size of $p = 40$ nodes fixed and varying T , though here with the effective signal-to-noise ratio $d/\sqrt{p \log T}$ as the ordinate (x -axis) rather than the scale factor d . As in Figure 4, the results of Theorems 1 and 2 are confirmed, with the error decaying quickly and also accurate recovery even at signal-to-noise ratios

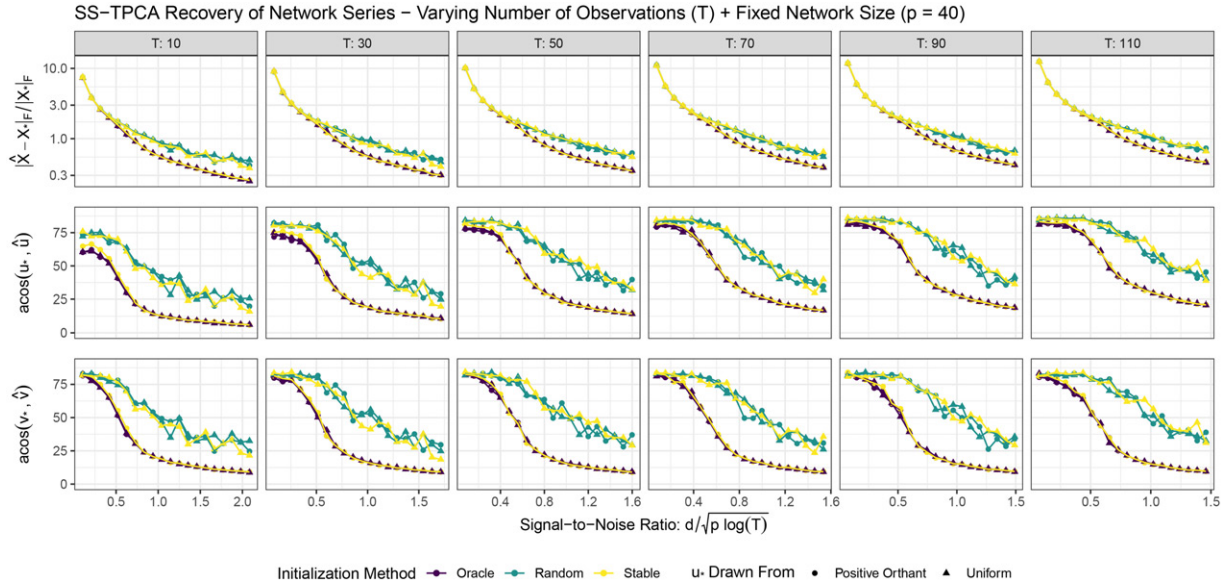


Figure 5. Performance of [Algorithm 1](#) for recovery of a rank-1 semi-symmetric tensor of norm d with added noise from a Gaussian Orthogonal Ensemble. Under the same generative model as [Figure 4](#), we here fix the dimension of V_* at 40 and vary the number of observations from $T = 10$ (far left) to $T = 110$ (far right). Here we rescale the x -axis to the effective signal-to-noise ratio $d/(\sqrt{p} \log T)$ and note that accuracy to within 25° can be obtained even in relatively difficult $\text{SNR} \approx 1$ scenarios. For $p = 40$, $\angle(\hat{V}, V_*) \approx 25^\circ$ corresponds to an ℓ_2 error of $\sqrt{2} \sin(25^\circ) \approx 0.6$ or a root mean squared error 0.1 across all elements of $\hat{V}$. This is well within the bounds of usability for many network science applications where the elements of $\mathcal{X}$ take $\{0, 1\}$ -values.

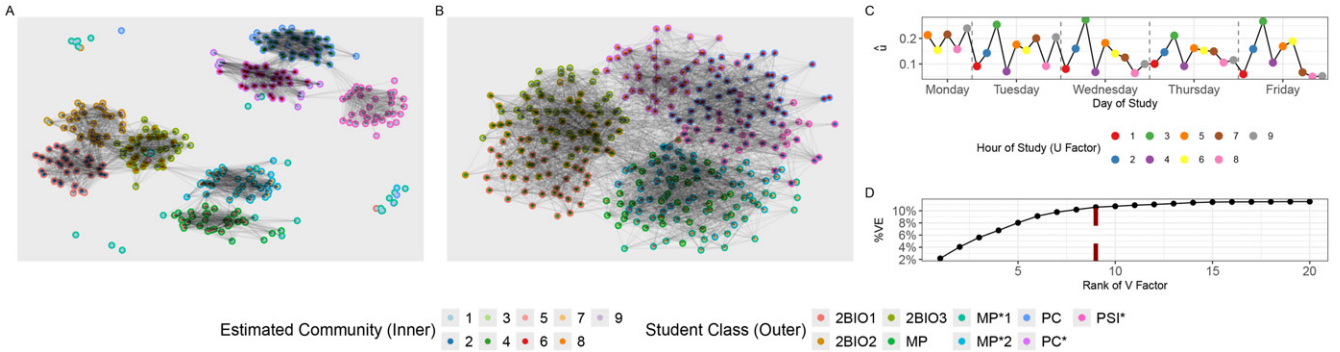


Figure 6. Application of SST-PCA to the French High School data described in [Section 4.2](#). Panel A depicts the SST-PCA principal network $\hat{V} \circ \hat{V}$ and the results of community detection performed on the principal network (inner color depicts estimated communities; outer ring depicts true communities). Eight of the estimated communities align with the true labels, while the final largest community (8, orange) captures low-degree vertices from all classes. SST-PCA also provides a time factor, $\hat{u}$, depicted in Panel C; examining $\hat{u}$, we see that the principal network structure is relatively constant over time, consistent with the different student specializations, but that the signal is clearest in the third hour of each study day (green dots). Panel D depicts the percentage of variance explained (%VE) by single-factor SST-PCA as a function of the rank of the principal network, with ranks greater than nine providing only marginal improvement. Finally, Panel B depicts the results of community detection on an aggregated network, formed by linking nodes which share an edge at any time point. Compared with the principal network, the aggregated network is far more dense and does not exhibit as many clear communities, though it avoids difficulties associated with the low-degree nodes of the principal network. Data made available by the SocioPatterns collaboration (sociopatterns.org).

of approximately 1. Further, both random and stable initializations are qualitative competitive with oracle initialization and with the stable initialization being essentially indistinguishable from the oracle scheme for positive u_* . Taken together, these two experiments demonstrate that [Algorithm 1](#) is robust to the initialization scheme and consistently performs well, even in settings where our initialization and signal strength assumptions are violated.

Additional simulation studies in Section F of the supplementary materials demonstrate Multi-Factor ($K > 1$) SST-PCA, demonstrate robustness of SST-PCA to moderate misspecification, and examine data-driven approaches to selecting both K and r .

4.2. Case Study: High School Contact Patterns

Finally, we apply SST-PCA to analyze the social dynamics of $p = 327$ high-school students from Marseilles, France using data originally collected by Mastrandrea, Fournet, and Barrat (2015) over five days in December 2013. They equipped students with wearable proximity sensors and recorded a total of 188,508 face-to-face interactions between students at a twenty second temporal resolution throughout the school day. They additionally collected the students' self-reported social contacts and selected metadata, including the students' class enrollments and genders; we do not use these node-specific covariates in estimation, but we use the class enrollment data to validate community detection results.

We reduce this data to $T = 41$ separate networks, each corresponding to a particular hour of school (e.g., Wednesday 2:00p.m.–3:00p.m.) by creating an edge if there was a recorded interaction between a pair of students during that window; repeated interactions during the same hour were discarded, creating distinct undirected unweighted graphs on 327 nodes, which we arrange into a semi-symmetric tensor $\mathcal{X} \in \mathbb{R}^{327 \times 327 \times 41}$. The resulting networks have average degree 3.05 ± 0.45 with a maximum degree of 30. We note that these networks are *highly sparse*: within each single network, approximately 30% of students have no recorded interactions, yielding 100 unconnected (degree zero) vertices at each time point. Over our entire sample, each student has at least 4 recorded interactions, so there are no persistently isolated vertices. Furthermore, there are very few persistent edges across time: of all possible edges, only 0.1% (79 out of 53,301) appear in more than half of the individual networks.

We focus our analysis on the second SST-PCA factor, as the first SST-PCA factor is approximately the sample mean for this data and deflation is essentially centering the data before performing PCA. Specifically, we apply the classical (Hotelling) deflation scheme described in Section 2.1 and then apply SST-PCA again to the resulting residuals. Because Hotelling deflation (or indeed any deflation) does not give exact zeros in the residual tensor, we rounded values to the nearest integer (0 or 1) before re-applying SST-PCA to the residuals. In order to select the rank r of the V -factor, we use the BIC selection criterion detailed in Section E.1 of the supplementary materials, which estimates that a rank $r = 9$ principal network will suffice for this data. Intuitively, these selections are reasonable, as student contacts during the school day are likely to be dominated by which of 9 “tracks” each student is assigned to (physical sciences, biological sciences, etc.). We fit single-factor SST-PCA to $\mathcal{X}$ with $r = 9$ using the stable initialization scheme described above. Figure 6 depicts the resulting principal network ($\hat{V} \circ \hat{V}$), time factor ($\hat{u}$), and the BIC used to select r .

To validate our findings, we apply the spectral clustering procedure of Newman (2006) on $\hat{V} \circ \hat{V}$ to divide students into 9 clusters, accurately recovering ground truth communities from the principal network. Furthermore, we see that the principal network ($\hat{V} \circ \hat{V}$) exhibits additional interesting higher-order structure, reflecting the fact that the nine tracks can be grouped into “super-track” which correspond to the topic of specialization. Specifically, we see that the three biology-focused tracks (2BIO1, 2BIO2, and 2BIO3) cluster together, as do the non-biology science and engineering tracks (PC, PC*, PSI). The mathematics focused tracks (MP, MP*1, MP*2) are somewhat less clear than the others, though the higher cluster of mathematics-tracked students is still clear.

As noted above, the individual networks have a large number of isolated vertices, so typical single network methods would require discarding almost a third of the graph before analysis. Given this sparsity, it is reasonable to consider temporal aggregation as suggested by Pensky and Zhang (2019): if we construct a graph with edges indicating *any* recorded interactions, we obtain a much denser graph, with average degree 35 and maximum degree 87. We can then compare the accuracy of communities estimated by eigenvector community detection on

the temporally aggregated network and the SST-PCA principal network. Using the students’ academic tracks as ground truth communities, the SST-PCA approach achieves an adjusted Rand index (ARI) of 89.5%, while the aggregated network yields an ARI of 67%. In addition to this improvement in community detection using the SST-PCA V factor, we can use the u factor to understand the temporal dynamics of this data: as shown in Figure 6, the u factor is relatively constant over time, but exhibits consistent spikes at the same hour of each day, reflecting a flexible period when students are allowed to have the most interactions with their classmates. This time factor lends itself to straightforward interpretation, in contrast to some competing methods, which may yield results that are less immediately interpretable. The u factor can be alternatively interpreted as a set of optimized weights used to estimate the principal network with a weighted average of observations; we discuss this viewpoint further, and the interpretation of $\hat{u}$ from this example in the supplementary materials.

Section F.1 of the supplementary materials provides additional details of this case study, including analysis of the first SST-PCA component, interpretation of the time factor $\hat{u}$, and comparisons with other approaches. Additional case studies in Section F.2 demonstrate the application of SST-PCA to judicial voting data and to correlation patterns of financial markets, including both EDA and SST-PCA-based changepoint detection.

5. Discussion

We have presented a novel framework for analysis of network-valued data based on a semi-symmetric tensor decomposition. A major contribution is the theoretical analysis of SST-PCA, where we show that SST-PCA achieves estimation accuracy comparable to classical PCA, despite the significantly higher dimensionality of the tensor problem. We also highlight connections between SST-PCA and a novel form of regularized PCA, based on a “rank-unvec” constraint. This constraint naturally captures a wide range of random graph models and can be applied in other supervised and unsupervised learning problems. Our analysis has three attractive properties: (i) it relies only on elementary tools of matrix perturbation theory; (ii) it establishes consistency under a very general noise model, allowing for the possibility of adversarial noise; and (iii) by analyzing a specific computational approach, it avoids considerations of global versus local optimality. Given this combination of ease and power, we expect that our analytic approach will find use elsewhere. We demonstrate through simulation that our method is robust to choice of initialization and outperforms other methods in low- and high-signal regimes. Finally, we demonstrate the effectiveness and interpretability of SST-PCA on a case study of high school contact patterns; additional details and two more case studies can be found in Section F of the supplementary materials. Code reproducing these analyses may be found at <https://github.com/michaelweylant/SSTPCA-Reproducibility>.

We expect several extensions of our work to be of particular interest: given the large scale of much network data, streaming, parallel, and approximate versions of Algorithm 1 will broaden the applicability of SST-PCA. Network time series also are

typically highly autocorrelated and regularization of the u -term reflecting this fact may significantly improve performance. While the low-rank model at the heart of SST-PCA captures a wide range of useful graph models, analogues of our work for directed, preferential attachment, multi-edge, and other graph structures would also be quite helpful in certain application domains. Robust versions of SST-PCA, building on similar techniques for matrix PCA, are also of interest. Finally, we have assumed that the set of vertices is fixed and consistent node labels are available: unfortunately, this is rarely the case for large-scale networks arising from social media, telecommunications, or other important domains. Node sets change from one day to the next as users create and delete accounts and, despite recent progress, the graph alignment problems resulting from unlabeled nodes remain computationally prohibitive.

Supplementary Materials

Supplementary materials include: a discussion of alternate approaches to multi-factor SST-PCA (Section A); an extension to functional SST-PCA (Section B); an elementary analysis of classical PCA (Section C); detailed proofs of Theorems 1 and 2 (Section D); methods for hyperparameter selection (Section E); additional empirical results (Section F); and further connections to related work (Section G).

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ORCID

Michael Weylandt  <http://orcid.org/0000-0002-0950-3881>

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